Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 2

Part A $C_xH_yO_x \rightarrow$

i) $\frac{2.883 g H_2O}{} \left| \frac{1 mol}{18.016 g} \right. = 0.1600244 \approx 0.1600 mol H_2O$

ii) The EF should be CHO because it's a 1:1 ratio meaning they're equal the divide by the smallest which gives CO_2 to 1.5 so $C_{1.5}HO_{1.5}$

Part B

i) $K_c = \frac{[H_3O^+][A_{sc}^-]}{[HA_{sc}]}$

ii) pKa = 1 $-\log\left(\frac{1}{10}\right)$

iii) the value is around $\frac{12}{2.5}$ or 13.

Question 2 is continued on the next page.

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Question 2

Part C

i) It is first order with respect to $[HA_{sc}]$ because in trials 1 & 3 when $[HA_{sc}]$ doubles & $[I_3^-]$ remains constant, the initial rate also doubles.

ii) rate $= K[HA_{sc}][I_3^-]$; $\ln[A]_t - \ln[A]_0 = -kt$

$K = 1/[HA_{sc}][I_3^-]$

Part D

LDF & D-D where I_3^- have a neg charge to attract H_2O.

Continue to Question 3.